A Prior-Odds Adjustment for the Probability of Direction in Multiple Testing

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Code is openly available in the *multibayes* repository on GitHub at https://github.com/mar-cald/multibayes.

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Abstract

Psychologists increasingly report the probability of direction (pd), the posterior probability that a parameter lies on the side of the null value favored by the posterior, and claim a direction when pd reaches a fixed threshold such as .975. Applied to many parameters, this rule produces more wrong claims as tests accumulate. We present an adjustment based on the classical Bayesian response to multiplicity, a penalty on the prior odds of each claim. Because the claimed direction is known only after the data, the penalty must cover both directions, so the researcher sets in advance $\bar{\pi}$, an upper bound on the prior probability of each direction of a parameter. A claim is made only if its posterior probability still reaches the threshold when its direction receives prior probability $1 - \bar{\pi}$, which is equivalent to a higher threshold for pd. The adjustment uses only the pd values and holds whenever the prior used to fit the model is continuous and symmetric about the null value. For exchangeable parameters, $\bar{\pi}$ bounds the expected proportion of parameters for which a directional claim would be false. Null effects are among them, so their expected proportion is a lower limit for $\bar{\pi}$. Simulations show how the false discovery rate, the family-wise error rate, and power depend on $\bar{\pi}$, set below, at, or above the proportion of null effects. We also compare the rule with two alternative procedures, joint directional statements and simultaneous credible intervals. All three are implemented in the accompanying R package *multibayes*.

Keywords: probability of direction, multiple testing, prior odds, FDR

A Prior-Odds Adjustment for the Probability of Direction in Multiple Testing

Over the past two decades, the use of Bayesian methods in psychological research has grown considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; van de Schoot et al., 2017). This expansion has, however, been accompanied by practices reminiscent of the "null ritual" in frequentist statistics, where widespread adoption gave way to automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). Researchers sometimes treat Bayes factor thresholds (e.g., BF > 3) as dichotomous decision rules, or claim that an effect "exists" because the posterior probability of a directional effect exceeds a conventional benchmark such as .975 (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). Such cutoffs are convenient, but they encourage all-or-none interpretations. Moreover, when the same cutoff is applied to parameter after parameter, the resulting decisions face the multiplicity problem long recognized in frequentist statistics.

Frequentists developed methods to address this problem, controlling either the Family-Wise Error Rate (FWER; the probability of at least one false positive) or the False Discovery Rate (FDR; the expected proportion of false positives among rejected hypotheses) (Benjamini & Hochberg, 1995; Westfall & Young, 1993). These procedures address whether a spurious result enters the literature, but leave unresolved a subtler consequence of the same selection process. When researchers examine many parameters but highlight only the most extreme estimates, selection systematically inflates reported magnitudes (Altoè et al., 2020; Gelman & Carlin, 2014; van Zwet & Cator, 2021). Bayesian methods can mitigate this overestimation, pulling extreme estimates toward more plausible values through informative priors (Dawid, 1994; van Zwet & Gelman, 2022) and hierarchical modeling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011). Magnitude bias and false-positive accumulation are, however, distinct problems. Shrinkage offers no guarantee about the error properties of a decision rule applied across many parameters: if the same cutoff is used repeatedly (e.g., claiming an effect whenever a posterior probability exceeds .975), the probability of spurious findings can remain substantial. Bayesian responses to this issue do exist, such as joint

probability statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020) or priors over the hypothesis space that scale with the number of tests (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997).

Here we focus on a decision rule based on the probability of direction (pd), which is increasingly used in psychological research (Ben-Artzi et al., 2022; Bramley & Ruggeri, 2022; Brissenden et al., 2023, 2024; Ciranka et al., 2026; Hochman et al., 2025; Makowski et al., 2020; Schurr et al., 2024; Stone-Sabali et al., 2024). The pd is the posterior probability that a parameter lies in its dominant direction (Makowski et al., 2019; Shi & Yin, 2021), and a direction is claimed when pd exceeds a fixed threshold (e.g., .95 or .975). We examine the frequentist operating characteristics[1] of this rule and address the resulting multiplicity problem with the second strategy noted above, a penalty on the prior odds of each claim. This penalty is the classical Bayesian response to multiplicity, introduced by Jeffreys (1938) and developed by Westfall et al. (1997), and it underlies more recent Bayesian approaches to FDR control (Abraham et al., 2022; Efron et al., 2001; Guo & Heitjan, 2010; Scott & Berger, 2006; Scott & Berger, 2010; Storey, 2002, 2003) as well as the post hoc correction implemented in JASP (de Jong, 2019; van den Bergh et al., 2020).

To our knowledge, the penalty has not been applied to pd. The obstacle is that the direction of a claim based on pd is chosen by the data, so the penalty cannot be attached to a direction fixed in advance. Following robust Bayesian analysis (Berger, 1994), we therefore consider a set of priors that lower the prior odds of either direction, and accept a claim only if it holds under all of them, that is, if its posterior probability under the prior most skeptical of it reaches the usual threshold. This is equivalent to raising the threshold that pd must reach. The prior used to fit the model, which we call the estimation prior, is one member of this set. The model is fitted only once, under this prior, and the posterior probability of each direction under any other member of the set follows directly from pd. The size of the penalty is set by a single number chosen by the researcher before the data, $\bar{\pi}$, an upper bound on the prior probability of

[1] Although frequentist and Bayesian frameworks rest on different foundations, examining the frequentist operating characteristics of a Bayesian procedure (how it behaves over repeated use) is a well-established practice (Müller et al., 2004; Ventz & Trippa, 2015; Zhang et al., 2019).

each direction of a parameter. When the parameters in a family are exchangeable, that is, when no parameter is judged more likely than another to lie on a given side of the null value, $\bar{\pi}$ also bounds the expected proportion of parameters for which a directional claim would be false. Null effects are among them, so their expected proportion is a lower limit for $\bar{\pi}$. This lower limit can be used as the value of $\bar{\pi}$ when the data are informative enough that a non-null effect is unlikely to be claimed with the wrong sign.

The paper proceeds as follows. We first introduce pd and the rule based on it, and show why the rule produces more false directional claims as the number of tests grows. We then derive the adjustment, which needs only the pd values and amounts to a higher threshold for pd, and discuss how to choose $\bar{\pi}$. A simulation study then shows how the FDR, the FWER, and power depend on $\bar{\pi}$, including values below and above the actual proportion of null effects, and compares the rule with joint directional statements and simultaneous credible intervals. Finally, we illustrate with a worked example the R package `multibayes`, which implements all three procedures, and conclude with a general discussion.

Probability of Direction

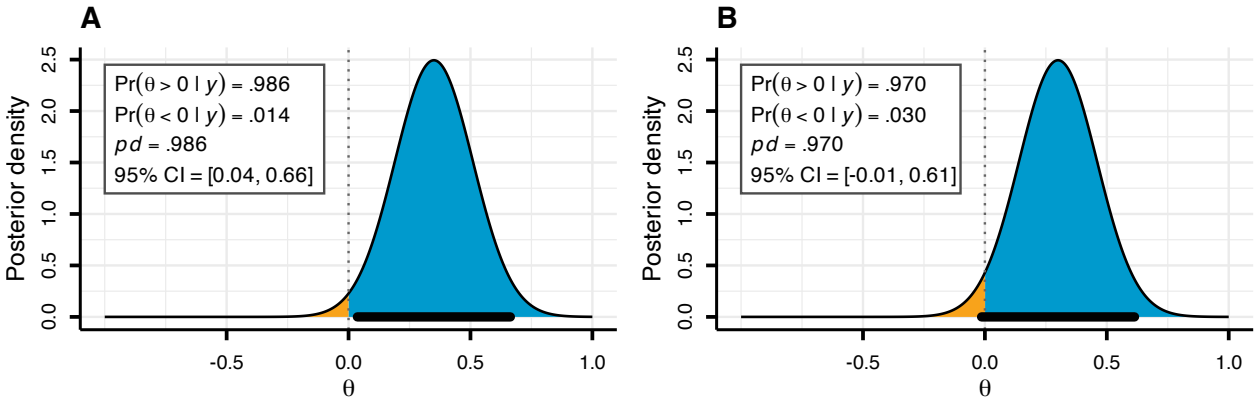
The probability of direction (pd) is a posterior summary that quantifies how strongly the posterior distribution favors one direction of a parameter θ relative to a reference value θ_{null} (Makowski et al., 2019). For a continuous posterior, it can be defined as the larger of the two one-sided posterior probabilities, $pd = \max(\Pr(\theta < \theta_{\text{null}} \mid y),\ \Pr(\theta > \theta_{\text{null}} \mid y))$, and it ranges from .5 (no directional preference) to 1 (all on one side). The pd equals exactly the complement of the one-sided p-value when computed on the posterior of a location parameter in exponential-family models estimated under a uniform prior; under a broader class of priors, this correspondence holds approximately (Casella & Berger, 1987; Marsman & Wagenmakers, 2017; Shi & Yin, 2021). For ease of exposition, we set $\theta_{\text{null}} = 0$ throughout; it can take any other value, which is particularly relevant when hypotheses involve a minimum effect of practical interest.

In practice, pd is often used to summarize both the direction and the existence of an effect. This use is clearest when phrased as a decision rule, $pd \geq c$, which requires the posterior to assign

at least c probability mass to the side of 0 it favors. Setting $c = 1 - \alpha/2$ is convenient because it bounds the posterior mass on the other side of 0 by $\alpha/2$. For concreteness, suppose the posterior favors positive values, so that $pd = \Pr(\theta > 0 \mid y)$. With $\alpha = .05$, $c = .975$, and the rule requires $\Pr(\theta \leq 0 \mid y) \leq .025$. The rule $pd \geq .975$ is therefore the same as the common practice of claiming an effect when the central 95% equal-tailed credible interval does not extend past 0 (Figure 1), which is often read as showing that 0 is relatively implausible (Box & Tiao, 1992). Like the reference value of 0, the threshold $c = .975$ is chosen only for ease of exposition, and any other posterior probability cutoff can be used.

Figure 1

The Probability of Direction and the 95% Credible Interval



Note. Two normal posteriors for a parameter θ, with the posterior mass below 0 in orange and above 0 in blue; the black bar is the central 95% equal-tailed credible interval. Panel A: $pd \geq .975$, and the interval does not extend past 0. Panel B: $pd < .975$, and the interval includes 0.

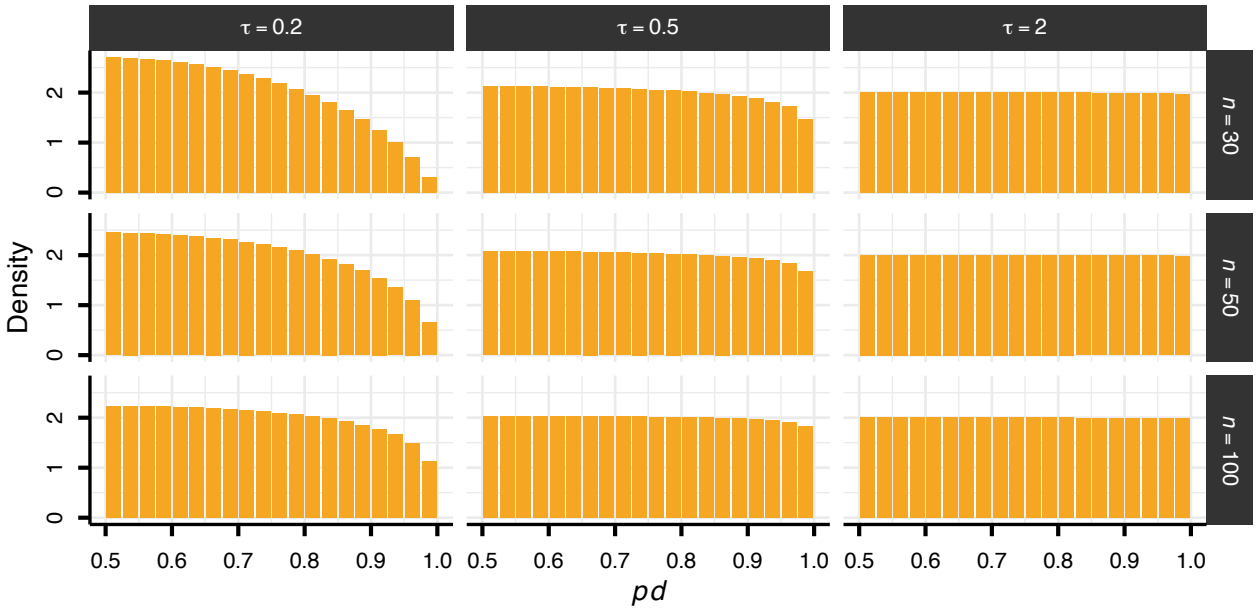
How often the rule $pd \geq c$ claims a direction when $\theta = 0$ depends on how pd varies across repeated samples, and hence on the model and the prior. All illustrations and simulations use the normal model, as a representative case of parameters whose posterior is approximately normal. For a parameter θ, n independent observations are drawn from $\mathcal{N}(\theta, 1)$, and θ has the prior $\mathcal{N}(0, \tau)$, where τ is the prior standard deviation, so that the posterior of θ is also normal. Small values of τ give a narrow prior, concentrated near 0, and large values a diffuse one.

In this model, for a null effect ($\theta = 0$) under a diffuse prior, pd is approximately uniform on $[.5, 1]$ across repeated samples (Figure 2, $\tau = 2$). The rule $pd \geq c$ with $c = 1 - \alpha/2$ therefore claims a direction with probability about α. A narrow prior shrinks pd toward .5 and lowers this

probability (Figure 2, $\tau = 0.2$), but only for each test separately. With any fixed prior, the probability that at least one of $pd_1, \ldots, pd_m$ reaches c still grows with the number of tests m. For $m = 10$ independent tests of null effects with $n = 50$, for example, it is about .40 under the diffuse prior and .15 under the narrow one.

Figure 2

Distribution of the Probability of Direction Under the Null



Note. Density of pd across repeated samples when $\theta = 0$, for observations $y_i \sim \mathcal{N}(\theta, 1)$ and a normal prior $\theta \sim \mathcal{N}(0, \tau)$, by sample size n (rows) and prior standard deviation τ (columns). With a diffuse prior ($\tau = 2$), pd is approximately uniform on $[.5, 1]$; a narrow prior ($\tau = 0.2$) shrinks it toward .5, especially in small samples. Bin width = 0.025.

Penalizing the Prior Odds of a Directional Claim

To limit the false claims that accumulate across tests, we penalize the prior odds of each directional claim, following Jeffreys (1938) and Westfall et al. (1997).

From the Probability of Direction to the Bayes Factor

As in the previous section, suppose that the posterior favors positive values. The claim is then that $\theta > 0$, which we denote H_+, and the opposite direction, $\theta < 0$, we denote H_-. We call the prior on θ specified in the fitted model the *estimation prior*, because the parameters are estimated under it. We assume that it is continuous and symmetric about 0, as are normal or Student-t priors

centered on 0 and, as a limit, flat priors. Symmetry makes H_+ and H_- equally probable a priori, $\Pr(H_+) = \Pr(H_-) = .5$, and after the data $pd = \Pr(H_+ \mid y)$ and $1 - pd = \Pr(H_- \mid y)$.

The ratio $pd/(1 - pd)$ is therefore the posterior odds of H_+ against H_-, that is, how many times more probable H_+ is than H_- after the data. By Bayes' rule, the posterior odds are the product of two terms,

$$\frac{\Pr(H_+ \mid y)}{\Pr(H_- \mid y)} = \mathrm{BF}_{+-} \times \frac{\Pr(H_+)}{\Pr(H_-)}.$$

The second term, the prior odds, says how many times more probable H_+ is than H_- before the data. The first, the Bayes factor BF_{+-}, says by how much the data change these odds. Under the symmetric estimation prior, the prior odds are $.5/.5 = 1$, so the posterior odds equal the Bayes factor (Marsman & Wagenmakers, 2017),

$$\frac{pd}{1 - pd} = \mathrm{BF}_{+-}.$$

For a parameter with $pd = .988$, for example, $\mathrm{BF}_{+-} = .988/.012 \approx 82$, so the data are about 82 times more probable under H_+ than under H_-.

The rule $pd \geq c$ can thus be written as a rule on the Bayes factor. Because the odds $pd/(1 - pd)$ increase with pd,

$$pd \geq c \quad \Longleftrightarrow \quad \frac{pd}{1 - pd} \geq \frac{c}{1 - c} \quad \Longleftrightarrow \quad \mathrm{BF}_{+-} \geq \frac{c}{1 - c}.$$

With $c = .975$, a direction is claimed whenever the Bayes factor is at least $.975/.025 = 39$. Because the prior odds are fixed at one, the Bayes factor alone decides every claim.

Prior Odds With Many Tests

Consider now a family of m tests, indexed by $i = 1, \ldots, m$, each concerning a parameter θ_i with probability of direction pd_i. Under the symmetric estimation prior, each parameter is equally likely to be positive or negative before the data. Whichever direction is later claimed for a parameter, the claim is therefore false with prior probability .5. If the parameters are independent

a priori, the prior probability that all m claims are false, an event we denote H_0, is the product of these probabilities, $.5^m$. With $m = 5$, $\Pr(H_0) = .5^5 \approx .03$, so before any data the estimation prior gives probability $1 - .03 = .97$ to at least one claim in the family being correct. This probability approaches one as more parameters are tested, yet the rule $pd \geq c$ requires the same Bayes factor of 39 for every claim.

The classical response to this problem (Jeffreys, 1938; Westfall et al., 1997) is to lower the prior odds of each claim, so that a larger Bayes factor is needed before the claim is made. Suppose first that the direction of a parameter is predicted before the data, say $\theta > 0$. The researcher then gives the opposite direction, H_-, a prior probability π larger than .5, and H_+ the probability $1 - \pi$, so the prior odds of H_+ become $(1 - \pi)/\pi$, below one.

The prior odds can be lowered without changing the shape of the estimation prior on either side of 0. For the prior $\mathcal{N}(0, \tau)$ of the normal model, let $\mathcal{N}_-(0, \tau)$ and $\mathcal{N}_+(0, \tau)$ denote this normal truncated to negative and to positive values, so that the estimation prior is $.5\,\mathcal{N}_- + .5\,\mathcal{N}_+$. The reweighted prior

$$\pi\,\mathcal{N}_-(0, \tau) + (1 - \pi)\,\mathcal{N}_+(0, \tau)$$

moves probability from the positive to the negative side and keeps the shape within each side. The likelihood averaged over each side, and hence the Bayes factor $pd/(1 - pd)$, is the same as under the estimation prior. Only the prior odds change, and the posterior odds of H_+ become

$$\mathrm{BF}_{+-} \times \frac{1 - \pi}{\pi}.$$

The remaining question is how large π should be. Jeffreys (1938) and Westfall et al. (1997) fixed the prior probability of H_0 in advance, at the same value whatever the size of the family. With the penalty, $\Pr(H_0) = \pi^m$ for independent parameters, so keeping $\Pr(H_0)$ fixed requires π to grow with m. Fixing $\Pr(H_0) = .5$, for example, makes the event that all claims are false and the event that at least one claim is correct equally probable before the data, and gives $\pi = .5^{1/m}$, which is .71 for $m = 2$, .87 for $m = 5$, and .97 for $m = 20$. A larger π gives smaller

prior odds, so the Bayes factor needed for a claim grows with m, much as the per-test threshold does in the frequentist Bonferroni correction (Westfall et al., 1997).

A Set of Priors Penalizing Either Direction

With pd, however, the direction to be claimed is not known before the data, because it is the side the posterior favors. We therefore apply the penalty of the previous section to each direction in turn. With $\pi = .8$, for example, this gives two reweighted versions of the estimation prior, one that gives $\theta < 0$ probability .8 and so penalizes the claim $\theta > 0$, and one that gives $\theta > 0$ probability .8 and so penalizes the claim $\theta < 0$. The value .8 is treated as an upper bound, so every prior in between is also allowed, including the estimation prior with .5 on each side.

In general, the researcher fixes before the data an upper bound $\bar{\pi}$, between .5 and 1. The class Γ, the set of priors considered, contains every prior obtained by reweighting the two halves of the estimation prior so that each side of 0 has prior probability between $1 - \bar{\pi}$ and $\bar{\pi}$ (Figure 3 shows the class for $\bar{\pi} = .7$ and $\bar{\pi} = .9$). With $\bar{\pi} = .5$, the estimation prior is the only prior in Γ, and no penalty is applied.

A claim about θ_i is made only if its posterior probability is at least c under every prior in Γ. As in the previous section, these priors share the Bayes factor and differ only in the prior odds, so this probability is smallest under the prior that gives the claimed direction the lowest prior probability, $1 - \bar{\pi}$ (solid line in Figure 3 for the claim $\theta > 0$). We call it the prior most skeptical of the claim. Under this prior, the posterior odds of the claim are those of the previous section with $\bar{\pi}$ in place of π, and the corresponding probability is

$$pd_{\text{adj},i} = \frac{pd_i\,(1-\bar{\pi})}{pd_i\,(1-\bar{\pi}) + (1-pd_i)\,\bar{\pi}}.$$

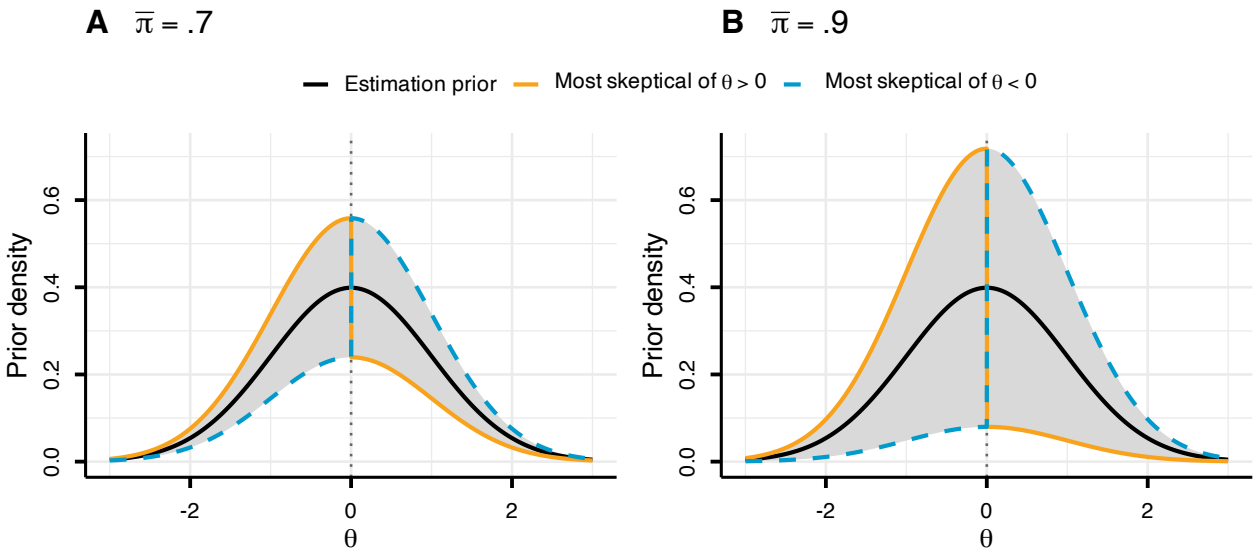
A claim is made when $pd_{\text{adj},i} \geq c$. Since $pd_{\text{adj},i}$ increases with pd_i, the same rule can be written as $pd_i \geq c^*$, with

$$c^* = \frac{c\,\bar{\pi}}{(1-c)(1-\bar{\pi}) + c\,\bar{\pi}}.$$

With $c = .975$ and $\bar{\pi} = .8$, $c^* = .9936$, which corresponds to a Bayes factor of 156 instead of 39.

Figure 3

The Class of Priors for Two Values of $\bar{\pi}$



Note. The priors in Γ reweight the two halves of the estimation prior, here $\mathcal{N}(0, 1)$ (black), keeping the shape of each half; every prior in Γ, including the estimation prior, lies within the grey band. Panel A: for $\bar{\pi} = .7$, the prior probability of $\theta > 0$ lies between .3 and .7; the colored lines are the two extreme priors of the class, the one most skeptical of $\theta > 0$ (solid) and the one most skeptical of $\theta < 0$ (dashed). Panel B: the same for $\bar{\pi} = .9$, for which the prior probability of $\theta > 0$ lies between .1 and .9. A larger $\bar{\pi}$ gives a wider class.

Since Γ is fixed before the data, the data decide only which direction is claimed, and hence which prior in Γ is the most skeptical.

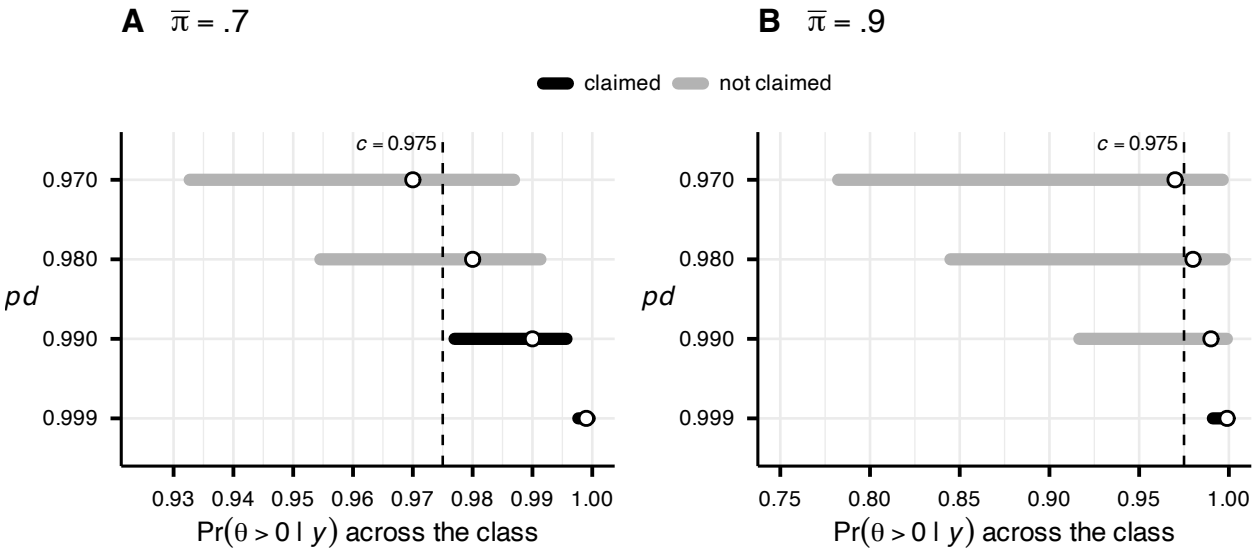
Unlike pd, which is at least .5 by definition, pd_{adj} can fall below .5. Take a parameter with $pd = .7$, favoring $\theta > 0$, and $\bar{\pi} = .8$. Its Bayes factor is $.7/.3 \approx 2.3$. Under the prior most skeptical of $\theta > 0$, the prior odds of $\theta > 0$ are .25 (.2/.8), so its posterior odds are $2.3 \times .25 \approx .58$ and $pd_{\text{adj}} \approx .37$. Under the prior most skeptical of $\theta < 0$, the prior odds of $\theta > 0$ are 4 instead (.8/.2), and its posterior probability is about .90. The priors in Γ thus disagree on which direction is more probable. A value of pd_{adj} below .5 does not mean that the data favor the opposite direction. It means that, with the skepticism allowed by $\bar{\pi}$, the data do not establish even which direction is more probable. In general, pd_{adj} falls below .5 whenever $pd < \bar{\pi}$.

Figure 4, Panel A, illustrates the rule for $\bar{\pi} = .7$, for which $c^* = .989$. Each bar spans the posterior probability of $\theta > 0$ across Γ, and its left end is pd_{adj}. A parameter with $pd = .980$ exceeds $c = .975$ under the estimation prior but not under every prior in Γ ($pd_{\text{adj}} = .955$), so no

claim is made, whereas parameters with $pd = .990$ and $pd = .999$ are claimed. With $\bar{\pi} = .9$ (Panel B), the class is wider (Figure 3, Panel B), and $c^* = .997$, so only the parameter with $pd = .999$ is claimed.

Figure 4

The Resulting Rule for Two Values of $\bar{\pi}$



Note. For four parameters, the range of the posterior probability of $\theta > 0$ across Γ (bars), whose left end is pd_{adj}, and its value under the estimation prior, pd (circles). A claim is made only when the whole bar lies at or above $c = .975$ (dashed line), that is, when $pd_{\text{adj}} \geq c$, which is equivalent to $pd \geq c^*$. Panel A: for $\bar{\pi} = .7$, $c^* = .989$. Panel B: for $\bar{\pi} = .9$, $c^* = .997$. A larger $\bar{\pi}$ gives a wider class and longer bars, so fewer parameters are claimed; note the different scales of the x-axes.

What $\bar{\pi}$ Means and How to Choose It

Like the estimation prior, every prior in Γ is continuous and gives null effects no probability. Null effects therefore enter the procedure not as prior mass but through the choice of $\bar{\pi}$, which sets how skeptical the priors in Γ may be. Besides bounding the prior probability of each direction of a parameter, $\bar{\pi}$ has a reading in terms of the family. Suppose the parameters are exchangeable, that is, no parameter is judged more likely than another to lie on a given side of 0, so the same $\bar{\pi}$ is applied to all of them. For exchangeable parameters, the prior probability that a parameter lies on a given side of 0 equals the expected proportion of parameters in the family that lie on that side, so $\bar{\pi}$ also bounds the expected proportion of parameters that are not on a given side. The priors in Γ give null effects no probability, but null effects are what make most

directional claims false. A claim that $\theta > 0$ is false if $\theta = 0$, and also if $\theta < 0$, that is, if a non-null effect is claimed with the wrong sign. A judgment about how large a proportion of the family could be null therefore gives a lower limit for $\bar{\pi}$. When the data are informative enough that wrong-sign claims are unlikely, this lower limit can be used as the value of $\bar{\pi}$. If up to 70% of the effects in a family could be null, $\bar{\pi} = .7$. When the data are less informative, wrong-sign claims become more likely, and a larger $\bar{\pi}$ is safer.

Alternatively, $\bar{\pi}$ can be set through a fixed $\Pr(H_0)$, as in Jeffreys (1938) and Westfall et al. (1997). For independent parameters, the largest value of $\Pr(H_0)$ over Γ is $\bar{\pi}^m$, so $\bar{\pi} = \Pr(H_0)^{1/m}$ grows with m. The first route keeps $\bar{\pi}$ the same whatever the size of the family and suits the aim of keeping the proportion of false claims low. The second suits the aim of avoiding any false claim.

Simulation Study

We conducted a fully crossed simulation study, with 10,000 replications per cell, varying four factors. The bound $\bar{\pi}$ set by the researcher took the values .5, .6, . . . , .9; the proportion of null effects in the data-generating process, π_0, took the values .5, .6, . . . , 1; the family size m was 4, 8, 10, or 20; and the sample size n was 30, 50, or 100.

At each replication the number of null effects ($\theta_i = 0$) was drawn from Binomial(m, π_0) and the remaining parameters were set to $\theta_i = 0.3$. The data-generating process thus places probability π_0 exactly at 0 and $1 - \pi_0$ at 0.3, whereas the priors in Γ are continuous. The simulation therefore shows whether a $\bar{\pi}$ set from the proportion of null effects gives decisions with good frequentist properties, even though the priors used to make them do not represent null effects. Data and posteriors followed the conjugate normal-normal model described in the section on pd, with n observations per parameter and a diffuse prior ($\tau = 2$), so that pd was computed exactly from the normal posterior.

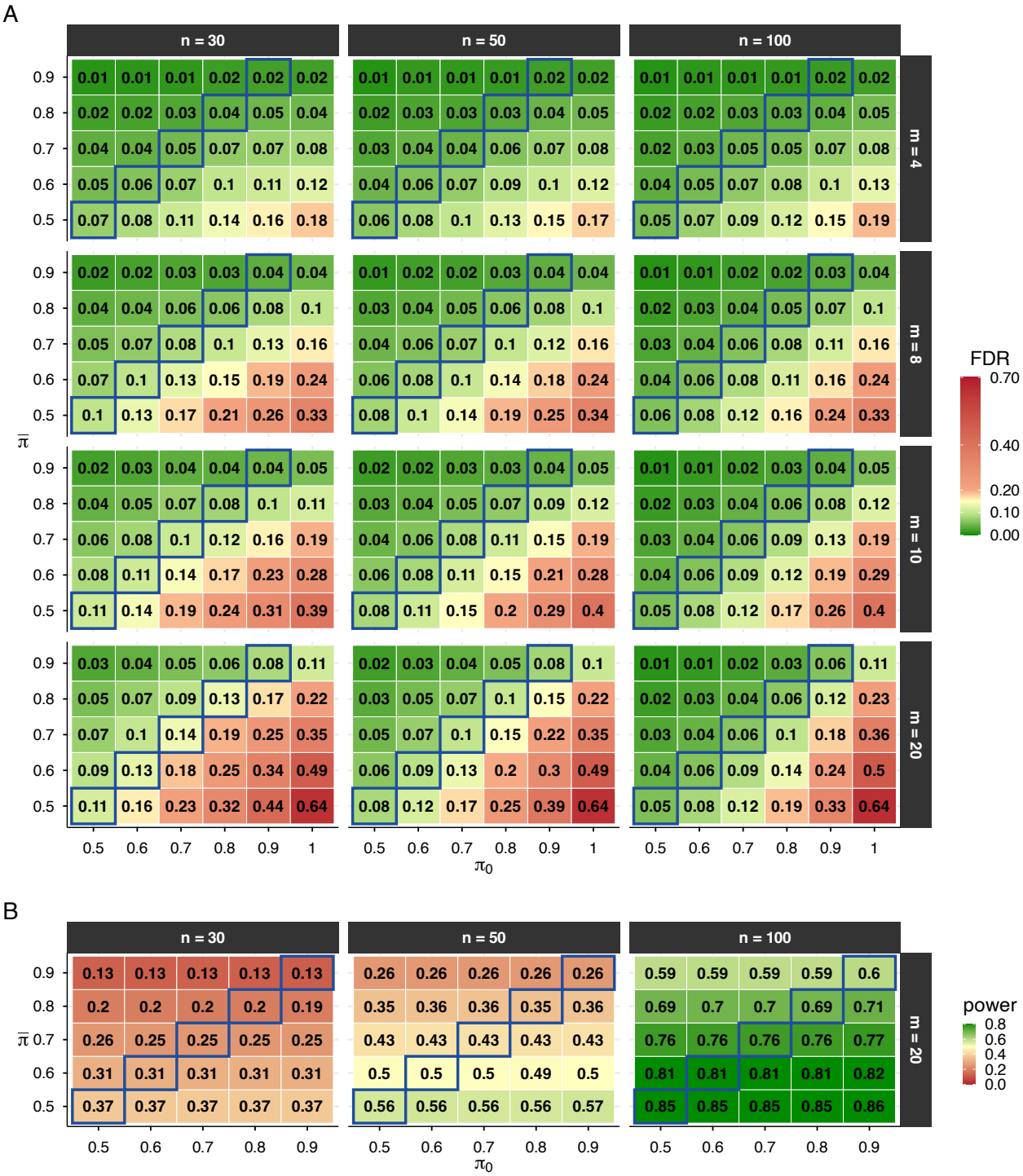
A directional claim was made when $pd \geq c^*$, with $c = .975$. For each family, let R be the number of claims and V the number of false claims. A claim about a null effect was false whichever direction it stated, whereas a claim about an effect of 0.3 was false only if it stated that $\theta_i < 0$, which requires a sample mean far below 0 and has probability at most .00015 in these

conditions. We evaluated the FDR, $\mathbb{E}[V/\max(R, 1)]$ (Benjamini & Hochberg, 1995; Benjamini & Yekutieli, 2005), the FWER, $\Pr(V > 0)$, and average power, the proportion of non-null effects claimed in the correct direction. The bound $\bar{\pi}$ plays the role of the researcher's judgment about how many null effects the family could contain, whereas π_0 is the proportion that the data-generating process actually produces on average. Since non-null effects were almost never claimed with the wrong sign, the lower limit given by the proportion of null effects is an appropriate value of $\bar{\pi}$ here. We refer to $\bar{\pi} = \pi_0$, a bound equal to the actual proportion of null effects, as a matched value, which corresponds to a correct judgment, to $\bar{\pi} < \pi_0$ as an underestimate, and to $\bar{\pi} > \pi_0$ as a conservative value.

Figure 5, Panel A, shows the FDR of the rule across the grid. In the row $\bar{\pi} = .5$, the unadjusted rule, the FDR inflates with the family size m and with π_0, reaching .64 when the family is large and most effects are null. Increasing $\bar{\pi}$ lowers the FDR. It stays low for matched and conservative values of $\bar{\pi}$, and rises only under substantial underestimation, when a small $\bar{\pi}$ is paired with a large π_0 at large m. Lowering the FDR in this way costs power, which decreases as $\bar{\pi}$ increases (Figure 5, Panel B). Average power depends only on $\bar{\pi}$ and n, not on m or π_0, because the threshold c^* is the same for every m and each non-null effect is claimed according to its own pd, so Panel B shows $m = 20$ only.

Figure 5

False Discovery Rate and Power of the Prior-Odds Rule

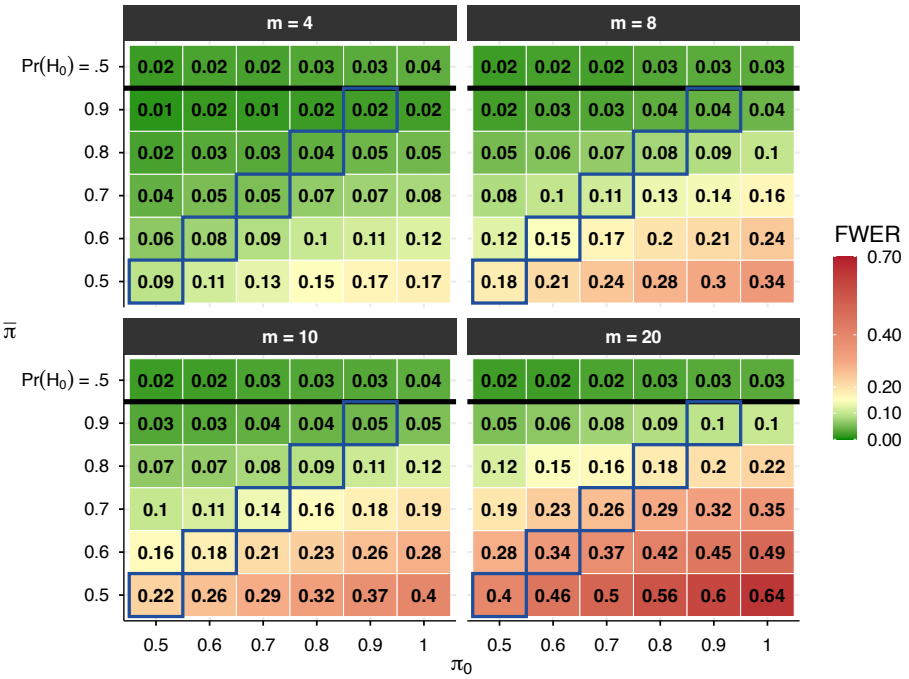


Note. False Discovery Rate (FDR, Panel A) and average power (Panel B) as a function of the true proportion of null effects, π_0 (x-axis), and the bound $\bar{\pi}$ set by the researcher (y-axis), by family size m (rows, Panel A) and sample size n (columns). The row $\bar{\pi} = .5$ is the unadjusted rule, where the prior odds are one and $c^* = c$; outlined cells mark matched values, $\bar{\pi} = \pi_0$. Each cell is the empirical FDR or power over 10,000 replications.

Figure 6 shows the FWER of the same rule (for $n = 50$, as a representative case). For a fixed $\bar{\pi}$ the FWER grows with the family size m, reaching .64 for the unadjusted rule with a large family of mostly null effects. Increasing $\bar{\pi}$ lowers it substantially. At $\bar{\pi} = .9$ it stays at or below about .10. Setting $\bar{\pi}$ through a fixed $\Pr(H_0)$ instead, which raises $\bar{\pi}$ with m, offsets this growth. For example, with a specified $\Pr(H_0) = .5$ (top row of Figure 6), the FWER stays at or below .04 for every family size.

Figure 6

Family-Wise Error Rate of the Prior-Odds Rule



Note. Family-wise error rate (FWER) as a function of the true proportion of null effects, π_0 (x-axis), and the bound $\bar{\pi}$ set by the researcher (y-axis), by family size m (panels), with $n = 50$. The row $\bar{\pi} = .5$ is the unadjusted rule; outlined cells mark matched values, $\bar{\pi} = \pi_0$. Each cell is the empirical FWER over 10,000 replications. With a fixed $\bar{\pi}$, the per-test threshold does not change with m, so the FWER is reduced by larger $\bar{\pi}$ but still grows with the family size m. The top row of each panel sets $\bar{\pi}$ through a specified $\Pr(H_0) = .5$, a choice of the researcher like the other values of $\bar{\pi}$, so that $\bar{\pi} = .5^{1/m}$.

Comparison With Other Bayesian Methods

We compare the prior-odds rule with two Bayesian alternatives. The first, hereafter the *joint directional rule*, orders the directional claims by their posterior probability and accepts the largest set whose joint posterior probability is at least c (Gelman & Tuerlinckx, 2000; Pruzek,

2016). The second, hereafter the *simultaneous-band rule*, computes credible intervals such that the joint posterior probability that every interval contains its parameter equals a set value, and claims a direction for a parameter whenever its interval excludes 0 (Box & Tiao, 1992; Schnell et al., 2016; Schnell et al., 2020).

The two procedures were applied to the same replications as the prior-odds rule, using 10,000 draws from each posterior, and we report the results for $n = 50$, as a representative case. The thresholds were set so that, for a single parameter, each procedure claims a direction when $pd \geq .975$. The joint directional rule requires a joint posterior probability of at least $c = .975$, the simultaneous band has a joint posterior probability of .95, and the prior-odds rule claims a direction when $pd_{\text{adj}} \geq c$. Figure 7 shows their FDR, FWER, and power, together with those of the prior-odds rule with a matched $\bar{\pi}$ ($\bar{\pi} = \pi_0$).

In this simulation setting, the FWER of the simultaneous-band rule ranges from .023 to .052, and its FDR is never larger. It therefore lies at or below .05, or within the Monte Carlo band around it, in all conditions. Its FWER increases with the proportion of null effects but hardly changes with m. Its power, however, decreases with family size.

The error rates of the joint directional rule, in contrast, increase with the proportion of null effects, and its FWER also with the family size. At $m = 20$, its FDR rises from .06 at $\pi_0 = .5$ to .37 at $\pi_0 = .9$ and .64 when all effects are null. Its first claim requires the same $pd \geq .975$ as the unadjusted rule (i.e., $\bar{\pi} = .5$), and each further claim must keep the joint probability at or above .975, so it makes a subset of the unadjusted rule's claims and never more false claims. When some effects are non-null, its FWER and FDR are lower than those of the unadjusted rule. When all effects are null, the two rules have practically equal error rates.

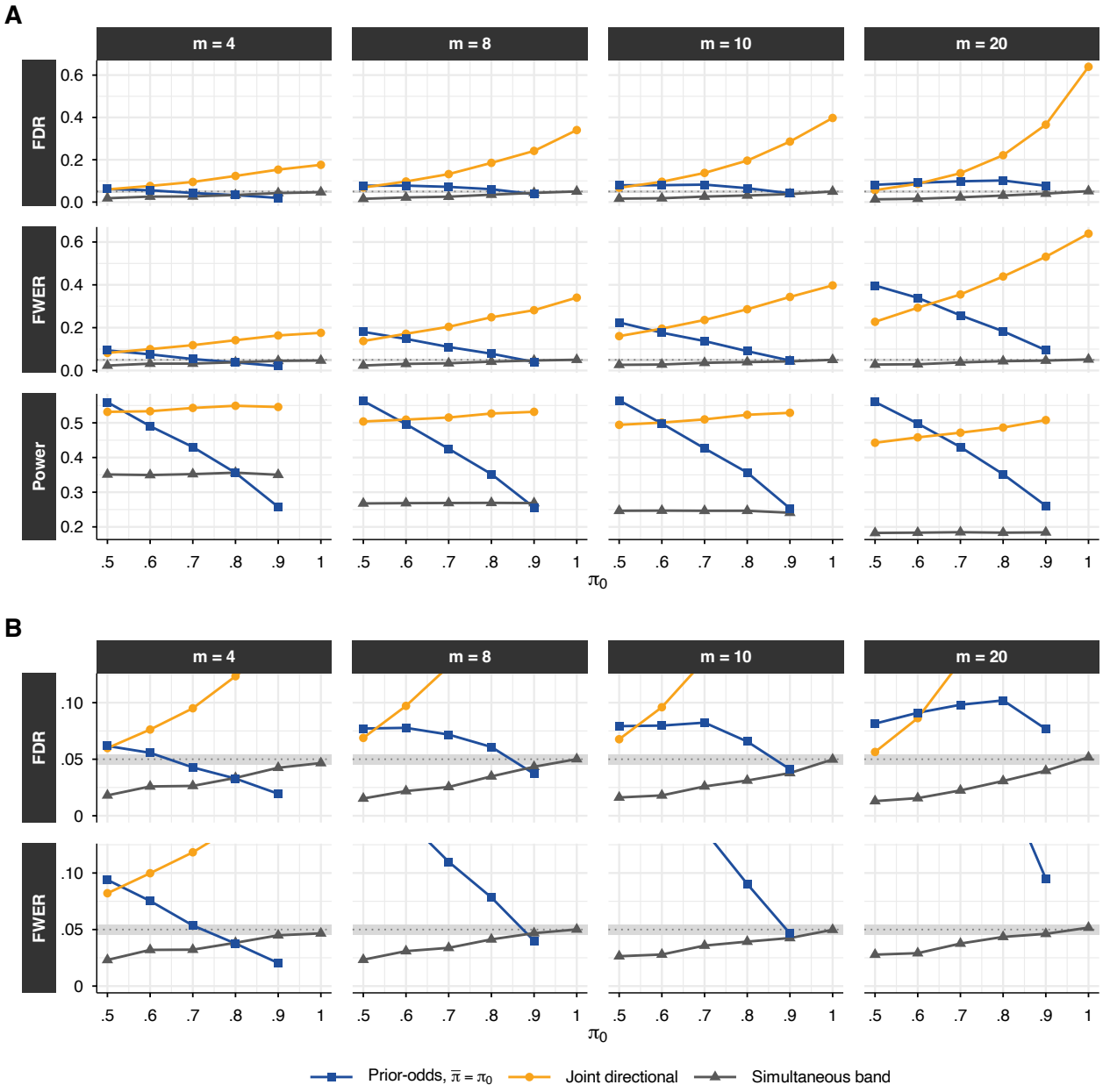
With a matched $\bar{\pi}$, the FDR of the prior-odds rule stays between .02 and .10, with the largest values at $m = 20$. It is therefore much lower than that of the unadjusted rule, but not always at or below .05, because the rule is not built to guarantee a nominal FDR. Its FWER decreases as π_0 increases, because a matched $\bar{\pi}$ increases with it. When most effects are null, its error rates are therefore far below those of the joint directional rule and approach those of the simultaneous band.

At $\pi_0 = .9$, for example, its FWER is at most .10, against up to .53 for the joint directional rule. Lower error rates come with lower power. The joint directional rule loses little power relative to the unadjusted rule, at most .03 at $m = 4$ and .12 at $m = 20$. The simultaneous band loses much more, and at $m = 20$ it has less power than the prior-odds rule with any $\bar{\pi}$. Unlike that of the other two procedures, the power of the prior-odds rule does not depend on m.

Which procedure to use thus depends on whether any false claim must be avoided (FWER) or only a small proportion of false claims is acceptable (FDR). The prior-odds rule is the only one of the three that requires an explicit specification of $\bar{\pi}$, and it is designed for the second aim. When any false claim must be avoided, the simultaneous band controlled the FWER directly in these simulations; the prior-odds rule can do so as well, with $\bar{\pi}$ set through a fixed $\Pr(H_0)$, but this is not its main purpose. The joint directional rule is better suited to settings in which the estimation prior describes the effects in the family, or in which a theory predicts the directions of a fixed set of parameters and the whole predicted directional pattern is evaluated as a single claim (i.e., not as a selected subset).

Figure 7

False Discovery Rate, Family-Wise Error Rate, and Power of the Three Procedures



Note. False Discovery Rate (FDR, top row), Family-Wise Error Rate (FWER, middle row), and average power (bottom row) as a function of the true proportion of null effects, π_0 (x-axis), by family size m (columns), with $n = 50$. Blue squares show the prior-odds rule with a matched $\bar{\pi}$, $\bar{\pi} = \pi_0$, which stops at $\pi_0 = .9$ because with $\bar{\pi} = 1$ no claim would ever be made, orange circles the joint directional rule (joint posterior probability of at least .975), and grey triangles the simultaneous-band rule (simultaneous credible intervals with joint posterior probability .95). The dotted line marks .05, and the grey band around it its 95% Monte Carlo band, $.05 \pm 1.96\sqrt{.05 \times .95/10{,}000}$, that is, .046 to .054. Panel A shows all rates on their full scale; Panel B shows the two error rates between 0 and .12, where the band is visible and points of the joint directional rule above .12 are cut off. Each point is the empirical rate over 10,000 replications.

Application: Worked Example With the R Package `multibayes`

The procedures are implemented in the `multibayes` R package. The function `pd.adjust()` accepts either a matrix of posterior draws, from which pd values are computed internally, or a vector of pre-computed pd values. The penalty is specified either directly as $\bar{\pi}$ (argument `pi.max`) or as $\Pr(H_0)$ (argument `p0`), from which $\bar{\pi} = \Pr(H_0)^{1/m}$. The decision threshold can be given as a pd cutoff c (argument `threshold`, default .975).

For this example, we use the openly available dataset from Lin et al. (2020), who investigated the effect of ego depletion on conflict resolution across four studies. In their paradigm, an initial high-demand (vs. low-demand) task was followed by a Stroop task. For simplicity, we focus on Study 1, which tested the effects of congruency and ego depletion (within subjects). The dataset is readily accessible via the `afex` package (Singmann et al., 2016) and was chosen for this reason; this example is not intended as a critique of the original work. Study 1 includes data from 253 participants, each completing 180 Stroop trials after each of the two tasks, and we use all of them.

```
# install.packages("remotes")
remotes::install_github("mar-cald/multibayes")
library(multibayes); library(brms); library(dplyr)

stroop1 <- subset(afex::stroop, subset = study == 1)
data_stroop <- stroop1 |>
  # remove useless levels id
  dplyr::mutate(id = droplevels(pno)) |>
  # select relevant variables
  dplyr::select(id, condition, congruency, trialnum, acc, rt) |>
  group_by(id, condition) |>
  mutate(prev_cong = lag(congruency)) |> # previous cong
  ungroup() |>
  na.omit()
```

Given the repeated-measures structure of the data and the large number of trials per participant, we fit a multilevel generalized linear model using `brms` (Bürkner, 2017) on reaction time (RT) data. A lognormal model was specified with the default flat priors. We used them

because they are the most common choice in applied work and the one in which the penalty matters most, not as a recommendation.

The model includes the effects of congruency (congruent vs. incongruent), previous congruency (congruent vs. incongruent), condition (depletion vs. control), and trial (position within the 180 trials of a condition, standardized). The sequential congruency effect was modeled as the interaction between congruency and previous congruency. A possible effect of depletion was modeled through condition and its interactions with congruency and trial, so that depletion could change response times overall, change the congruency effect, or change how either evolves over the trials.

The question of the study is whether ego depletion affects performance, so the family consists of the four effects involving condition (Table 1). The remaining effects, such as the congruency effect, are estimated but enter no claim, so they are not part of the family. Before the data we had no reason to judge any of the four more likely than another to be positive or negative, so we treat them as exchangeable and apply the same $\bar{\pi}$ to all of them. A family that also included the congruency effect, which is expected to be large and positive, would not be exchangeable.

Although a direction could be predicted for some of these effects (e.g., that depletion enlarges the congruency effect), we compute pd as $\max\big(\Pr(\theta > 0 \mid y), \Pr(\theta < 0 \mid y)\big)$ for all hypotheses. When the direction of each parameter is specified in advance, the argument `direction` (`"greater"` or `"less"`, for each hypothesis) computes pd as the posterior probability of the predicted direction, with the same threshold. The rule then corresponds to a single prior for each parameter, the member of Γ most skeptical of the predicted direction, and pd_{adj} is the posterior probability under that prior.

We removed trials without a response and, of the remaining trials, retained only correct trials preceded by a correct response (i.e., error and post-error trials were excluded), yielding 76,094 observations.

Table 1
Tested Hypotheses for Reaction Times

Hypothesis	θ_{null}
H_1: condition effect	0
H_2: congruency × condition interaction	0
H_3: condition × trial interaction	0
H_4: congruency × condition × trial interaction	0

```
data_rt <- data_stroop |>
  group_by(id) |>
  # remove error and post error trials
  filter(acc == 1 & lag(acc, default = 1) == 1) |>
  ungroup()

# set contrast
contrasts(data_rt$congruency) <- contr.sum(2)/2
contrasts(data_rt$prev_cong) <- contr.sum(2)/2
contrasts(data_rt$condition) <- contr.sum(2)/2
# trial standardized (z-score)
data_rt$trial <- as.numeric(scale(data_rt$trialnum))

mod_rt <- brm(
  formula = rt ~ 1 + congruency * prev_cong +
    congruency * condition * trial +
    (1 + prev_cong + congruency * condition + trial * condition || id),
  family = lognormal(), data = data_rt,
  cores = 3, chains = 3, iter = 8000, seed = 1)
```

Since `pd.adjust()` computes *pd* values automatically from a matrix of posterior draws, we extracted the draws of the four effects in the family and set $\bar{\pi} = .8$. Under the estimation prior, each sign of a parameter has prior probability .5; with $\bar{\pi} = .8$, we allow either direction of each parameter to have prior probability as high as .8, so a claim requires a Bayes factor four times larger than under the unadjusted rule. Because the four effects are exchangeable and the data are informative, this corresponds to judging that up to 80% of them, on average, could be null. This reflects skepticism about whether the depletion manipulation affects Stroop performance at all.

```r
draws_rt <- as.data.frame(mod_rt)
# the family: the four effects involving condition
par_names <- grep("^b_.*condition", colnames(draws_rt), value = TRUE)
draws_rt <- draws_rt[, par_names]
pd.adjust(draws = draws_rt, pi.max = 0.8, null.value = 0)
```

```
Prior-odds threshold for directional claims

  m = 4, pi.max = 0.8

  c = 0.975, c* = 0.9936

  null value = 0, direction = two.sided


                              median.est     pd pd.adj
b_condition1                      0.0064 0.8551 0.5960
b_congruency1:condition1         -0.0126 0.9986 0.9944
b_condition1:trial                0.0008 0.5881 0.2630
b_congruency1:condition1:trial    0.0026 0.7727 0.4594

pd.adj: Pr(favored direction | y) under the prior in the class most skeptical
        of it; 1 - pd.adj is the probability of the other side under that prior.
        Below 0.5, the priors in the class disagree on the more probable direction.
Claim a direction when pd.adj >= 0.975, or equivalently pd >= 0.9936.
```

The printed header reports m, $\bar{\pi}$, and the resulting thresholds: with the default $c = .975$, a directional claim requires $pd_{\text{adj}} \geq .975$ or, equivalently, $pd \geq c^* = .9936$. Each row gives the posterior median, pd, and pd_{adj} (`pd.adj`) of a parameter, and a short note below the table explains pd_{adj}.

Only the congruency × condition interaction (`b_congruency1:condition1`) has $pd \geq c$, with $pd = .9986$, and it also has $pd_{\text{adj}} = .994$, at or above c. The claim therefore holds under every prior in Γ. On the log scale, the congruency effect, the difference between incongruent and

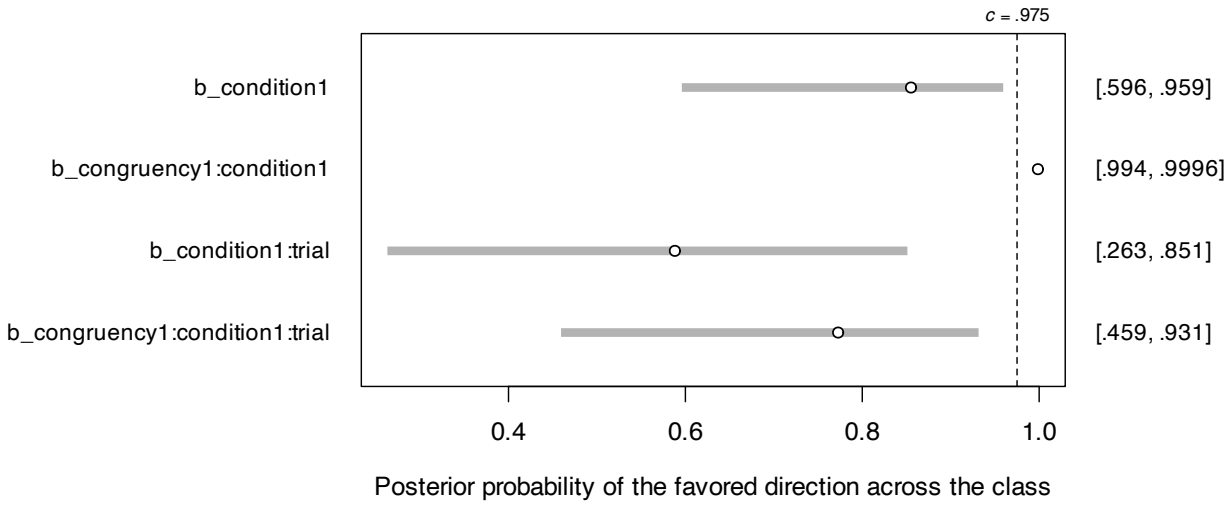
congruent trials, was .155 after the low-demand task and .143 after the high-demand task, so the claim is that the congruency effect is smaller after the high-demand task. None of the other three effects reaches c, with or without the adjustment. For two of them, the condition × trial interaction (pd = .588, pd_{adj} = .263) and the three-way interaction (pd = .773, pd_{adj} = .459), pd_{adj} is below .5, so the priors in Γ disagree on which direction is more probable, and the data do not establish even a leaning in either direction.

The function `plot()` draws, for each parameter, the range of the posterior probability of its favored direction across Γ (Figure 8):

```
plot(pd.adjust(draws = draws_rt, pi.max = 0.8))
```

Figure 8

Posterior Probability of the Favored Direction Across the Class for the Four Effects Involving Condition



Note. Output of `plot()` for the worked example ($\bar{\pi}$ = .8). Each bar spans the posterior probability of the favored direction across Γ, from pd_{adj} (left end) to its value under the prior most favorable to that direction (right end); points mark pd, and the two ends of each range are printed on the right. The dashed line marks c = .975. Only the congruency × condition interaction lies entirely at or above c. Its range, from .994 to .9996, is shorter than its point, because close to 1 the priors in Γ change the posterior probability very little. The bars of the condition × trial and three-way interactions cross .5.

A companion function, `joint()`, implements the simultaneous-band rule and the joint

directional rule compared in the simulation study. By default, it returns simultaneous equal-tailed credible intervals:

```
joint(draws_rt, prob = 0.95)
```

```
Simultaneous credible intervals (joint probability 0.95)

                                  lower     est   upper
b_condition1                    -0.0089  0.0064  0.0218
b_congruency1:condition1        -0.0229 -0.0126 -0.0023
b_condition1:trial              -0.0079  0.0008  0.0093
b_congruency1:condition1:trial -0.0060  0.0026  0.0113
```

Only the band of the congruency × condition interaction ($[-0.0229, -0.0023]$) excludes 0, so the simultaneous-band rule makes the same single claim.

Setting `interval = FALSE` returns instead the joint posterior probability of the k strongest directional claims, for each k:

```
joint(draws_rt, interval = FALSE, null.value = 0)
```

```
Joint directional statement
  null value = 0, direction = two.sided
  joint_cum: Pr(the top k claims all hold | y)

                               median.est     pd joint_cum
b_congruency1:condition1          -0.0126 0.9986    0.9986
b_condition1                       0.0064 0.8551    0.8538
b_congruency1:condition1:trial     0.0026 0.7727    0.6584
b_condition1:trial                 0.0008 0.5881    0.3668
```

The rows are sorted by decreasing pd, and `joint_cum` gives this probability for the first k rows. Here it is .9986 for the strongest claim alone and .85 for the two strongest, so the joint directional rule at .975 also accepts only the claim about the congruency × condition interaction. In this family, the three procedures agree.

Discussion

The probability of direction (pd) is the posterior mass on the data-favored side of a reference point null value, equivalently, the larger of the two one-sided posterior probabilities. It is common practice to claim a direction whenever pd exceeds a fixed threshold c, typically .975. Under an estimation prior symmetric about the null value, the direction selected after inspecting the data has the same prior probability as the opposite one, so each such data-selected hypothesis is evaluated at prior odds of one, and the rule $pd \geq c$ leaves these odds unchanged. When many parameters, independent a priori, are tested, these prior odds make it almost certain, before the data, that at least one favored direction is genuine. Building on Jeffreys (1938; see also Westfall et al., 1997) and on Bayesian approaches to FDR control (Abraham et al., 2022; Efron et al., 2001; Guo & Heitjan, 2010; Scott & Berger, 2006; Scott & Berger, 2010; Stephens, 2017; Storey, 2002, 2003), the procedure presented here requires each claim to hold even when its prior odds are lowered, by an amount the researcher sets through a single quantity, $\bar{\pi}$, an upper bound on the prior probability of each direction of a parameter. A claim is then made when pd_{adj}, its posterior probability under the prior most skeptical of it, reaches the usual threshold c, or, equivalently, when pd reaches the higher threshold c^*.

If the direction of a test is specified in advance, the rule reduces to a single prior for that parameter, the one that gives the predicted direction prior probability $1 - \bar{\pi}$. Whether or not the direction is specified in advance, the priors under which claims are evaluated are continuous and give point null effects no probability. Because the procedure evaluates only directional claims, not making a claim reflects the absence of a directional discovery, not evidence that the effect is null. Evaluating whether an effect is null requires other methods, such as those based on the two-group, or spike-and-slab, prior, which places probability exactly at the null value (Scott & Berger, 2006;

Scott & Berger, 2010; Stephens, 2017). In Bayesian approaches to FDR control based on this prior, the proportion of null effects can be learned from the data (Scott & Berger, 2010; Stephens, 2017), but this requires large families (Efron et al., 2001; Storey, 2002), larger than those typical of psychological studies.

The proposed adjustment keeps the researcher's continuous prior and accounts for null effects only in the decision. When the parameters in the family are exchangeable, $\bar{\pi}$ can be specified from a judgment about how many null effects the family could contain, instead of being learned from the data. Because a claim is also false when a non-null effect is claimed with the wrong sign, the proportion of null effects is a lower limit for $\bar{\pi}$. It can be used as the value of $\bar{\pi}$ when the data are informative enough to make wrong-sign claims unlikely, as in our simulations, whereas less informative data call for a larger value. In the simulations, the FDR increased gradually when $\bar{\pi}$ was below the proportion of null effects and decreased, at the cost of power, when it was above it. A researcher may also deliberately choose a $\bar{\pi}$ larger than the expected proportion of null effects, accepting this loss of power in exchange for fewer false claims.

The implementation applies directly to posterior draws or any posterior-derived pd, including cases where pd values arise from multiple separate analyses within the same manuscript (Blondé et al., 2021; Ciranka et al., 2026). This raises the question of how to define the hypothesis family to which the correction applies (Hochberg & Tamhane, 1987; Shaffer, 1995). Although no universal consensus exists regarding what constitutes a family, we favor the criteria proposed by Westfall et al. (2011), where a family is a set of questions that *"form a natural and coherent unit, [are] considered simultaneously in the decision-making process, and/or present a set over which one plays pick-the-winner"*. Regardless of the chosen definition, authors should clearly specify their chosen family and explain how it serves the study's substantive goals.

We further recommend that researchers justify their choice of $\bar{\pi}$ by drawing on theory, expert elicitation, or domain expertise (O'Hagan, 2019), and report this justification. Making this choice explicit serves two purposes. For researchers who use diffuse estimation priors, it provides a principled way to encode skepticism at the family level without placing informative priors on

individual parameters. For researchers who judge prior odds of one appropriate and decline the adjustment, it makes that commitment transparent.

A limitation is that applying the same $\bar{\pi}$ to every parameter assumes that they are exchangeable. In many models this is not the case; for example, a known main effect and an interaction term that motivated the study typically differ in prior plausibility. A common $\bar{\pi}$ set at the average proportion of null effects in the family keeps the overall error rate close to that of an exchangeable family, since the rule applies the same threshold to every parameter, but it distributes errors unevenly. The threshold is too strict for the more plausible claims, which lose power, and too lenient for the less plausible ones, among which false claims are more frequent. In such cases the family can be divided into more homogeneous subsets, each with its own $\bar{\pi}$, or a conservative common value can be used. Parameters on which no claim is based need not be included in the family.

The derivation of the adjustment requires an estimation prior that is continuous and symmetric about the null value, because only then is $pd/(1 - pd)$ the Bayes factor (Marsman & Wagenmakers, 2017). Informative priors, such as a narrow normal prior centered on the null value, meet this condition, as do weakly informative and flat ones. The error rates and power reported here, however, were obtained in the normal model with a diffuse prior, and may differ in other models and with other priors.

If a minimum effect of practical interest δ can be specified, the question changes from the direction of an effect to whether it lies outside the region $(-\delta, \delta)$ (Kruschke, 2018). This hypothesis is fixed before the data, so the classical penalty applies to it directly. With a Cauchy prior of scale δ, which places half its mass inside the region, the posterior odds equal the Bayes factor, and $\bar{\pi}$ is the prior probability that an effect is negligible. Unlike the adjustment proposed here, this approach needs a proper prior chosen in light of δ, and the direction of an effect outside the region is read from the posterior.

As Bayesian methods become increasingly common in psychological research, often with fixed decision thresholds, the implications of multiplicity should be explicitly recognized. The

adjustment proposed here asks for one additional number, $\bar{\pi}$, and can be applied to the pd values that researchers already report. We hope that claims based on pd will be reported together with the family to which they belong, the value of $\bar{\pi}$ used, including $\bar{\pi} = .5$ when no adjustment is made, and the pd and pd_{adj} of every parameter in the family.

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Declarations

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Conflicts of Interest/Competing Interests

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics Approval

Not applicable. The study reanalyzes openly available, anonymized data (Lin et al., 2020) and collected no new data.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Availability of Data and Materials

The worked example uses the openly available data of Lin et al. (2020), archived on OSF (Lin, 2020) and distributed with the `afex` R package (Singmann et al., 2016). The simulated data can be regenerated with the code described below.

Code Availability

The `multibayes` R package and the code to reproduce all simulations, figures, and the worked example are archived on Zenodo ([Zenodo DOI to be added]) and developed at https://github.com/mar-cald/multibayes.

Authors' Contributions

Author roles were classified using the Contributor Roles Taxonomy (CRediT; https://credit.niso.org/) as follows — Conceptualization: M. Calderan and G. Altoè; Methodology: M. Calderan, F. Gambarota, and L. Finos; Software: M. Calderan and F. Gambarota; Writing - Original Draft: M. Calderan; Writing - Review & Editing: F. Gambarota, L. Finos, and G. Altoè; Visualization: M. Calderan; Supervision: L. Finos and G. Altoè. All authors approved the final version of the manuscript and agree to be accountable for the work.

Open Practices Statement

The code to reproduce all simulations, figures, and the worked example is archived on Zenodo ([Zenodo DOI to be added]), and the analyzed data are available on OSF (Lin, 2020). The study was not preregistered.

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